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Assignment 6: Triage Pseudocode
Date: May 25, 2026

Triage Pseudocode (ABCDE-Based ED Risk Model)

This Python-style pseudocode implements an ABCDE-based triage framework loosely based on the Emergency Severity Index (ESI). It combines risk scoring, critical overrides, and multi-system escalation to assign ED triage levels (1–5), where level 1 indicates highest priority (critical patients who need immediate treatment). This model is intended for adult patients only; thresholds may vary depending on age and clinical context. For example, in cases where circulation or catastrophic hemorrhage is the primary concern, the assessment order may be modified to CAB (Circulation–Airway–Breathing) instead of ABCDE.

In this model, all vital sign values have been interpreted from established clinical reference ranges and early warning signs of patient deterioration, with supporting references listed below. Threshold values should not be interpreted as definitive clinical cutoffs. For this system to function correctly, patient vital sign inputs must be accurate and complete; however, the model includes handling for NaN or missing values to allow continued processing. It is designed as a decision-support system and not a substitute for professional clinical judgment.

A potential future application of this system would involve patients arriving with a QR containing their medical history, allowing the triage system to adjust accordingly with new vital sign readings to improve the accuracy of risk assessment.

Pseudocode:

```
# Input
Patient vital signs are received:
SpO2, RR, SBP, DBP, MAP, Pulse, GCS, Temp, Pain, Glucose

# Validation
IF critical values missing (SpO2, RR, SBP, Pulse, GCS):
      RETURN LEVEL 2 # precaution due to incomplete critical data

ELSE:
      continue with available values (exclude missing NaN values from scoring)

# INITIALIZE:
    RiskScore = 0

    BCount = 0
    CCount = 0
    DCount = 0
    ECount = 0
    EscalationLevel = 5 # lowest severity allowed initially
```

```
# A — AIRWAY
```

```
# Critical override — overrides all scoring
IF airway obstructed OR unable to speak OR any airway compromise:
```

```
      RETURN LEVEL 1 # Resuscitation
```

```
# B — BREATHING
```

```
# Critical override
IF SpO2 < 90 OR RR < 8 OR RR > 30:
      RETURN LEVEL 1

# Respiratory Rate (RR)
IF RR 12–20:
    score = 0
ELSE IF RR 8–11 OR 21–30:
    score = 1
ELSE:
    score = 2

RiskScore = RiskScore + score
IF score > 0: BCount = 1

# Oxygen Saturation (SpO2)
IF SpO2 >= 95:
    score = 0
ELSE IF SpO2 >= 90:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: BCount = 1
```

```
C — CIRCULATION
```

```
# Critical override
# DBP considered within scoring system; SBP and MAP used for critical
override.
IF SBP < 80 OR MAP < 50 OR Pulse < 40 OR Pulse > 140:
    RETURN LEVEL 1

# Limb Ischemia critical override
IF signs of severe limb ischemia or poor circulation
    (cold limb, pale or blue discoloration, mottled skin,
    absent pulse, or sudden severe limb pain with numbness or weakness):
    RETURN LEVEL 1

# SBP
IF 90–120:
    score = 0
ELSE IF 80–89 OR 140–180:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: CCount = 1
```

```
# DBP
IF 60-80:
    score = 0
ELSE IF 50-59 OR 90-120:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: CCount = 1

# MAP
IF 70-100:
    score = 0
ELSE IF 60-69 OR 101-110:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: CCount = 1

# Heart Rate (Pulse)
IF 60-100:
    score = 0
ELSE IF 50-59 OR 101-130:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: CCount = 1
```

D — DISABILITY

```
# Critical override
IF GCS ≤ 8 OR unresponsive:
    RETURN LEVEL 1
# GCS
IF GCS = 15:
    score = 0
ELSE IF GCS 13-14:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: DCount = 1
```

E — EXPOSURE

```
# Temperature (°C)
IF Temp 36.0-37.5:
    score = 0
```

```
ELSE IF Temp 35.0–35.9 OR 37.6–39.0:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: ECount = 1
```

 # Pain

```
IF Pain <= 3:
    score = 0
ELSE IF Pain 4–7:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: ECount = 1
```

 # Glucose

```
IF 4.0–7.0:
    score = 0
ELSE IF 3.0–3.9 OR 7.1–11:
    score = 1
ELSE:
    score = 2

RiskScore += score
IF score > 0: ECount = 1
```

---

6 — ESCALATION (based on number of affected systems ABCDE)

```
SystemCount = BCount + CCount + DCount + ECount

IF SystemCount ≥ 2:
      EscalationLevel = 2

IF SystemCount ≥ 3:
      EscalationLevel = 1
```

---

7 — SEPSIS EARLY WARNING RULE

```
IF Temp > 39°C
AND (Pulse > 120 OR RR > 22)
AND (SBP < 100 OR MAP < 65):

    Alert = "SEPSIS SUSPECTED"

    IF SBP < 90 OR MAP < 60:
        EscalationLevel = 1 # septic shock
    ELSE:
        EscalationLevel = 2 # sepsis without shock
```

---

8 — FINAL TRIAGE CLASSIFICATION

```
IF RiskScore >= 8:
    LEVEL = 1 # Resuscitation

ELSE IF 5 <= RiskScore <= 7:
    LEVEL = 2 # Emergency

ELSE IF 3 <= RiskScore <= 4:
    LEVEL = 3 # Urgent

ELSE IF 1 <= RiskScore <= 2:
    LEVEL = 4 # Semi-Urgent

ELSE:
    LEVEL = 5 # Non-Urgent

# Safety escalation constraint
IF EscalationLevel < LEVEL:
    LEVEL = EscalationLevel

RETURN LEVEL
```

9 — OUTPUT

```
# Outputs final triage level along with key indicators and the major clinical
issue driving escalation

RETURN:
    - LEVEL - Final Triage Level (1–5)      # Urgency level
    - RiskScore                             # Cumulative risk value
    - SystemCount           # Number of affected systems (B, C, D, E)
    - Major Issue           # Primary concern(e.g.Breathing,Circulation)
    - ABCDE override triggered          # Critical override triggered (Yes/No)
    - Sepsis Alert (if triggered)       # Sepsis criteria triggered (Yes/No)
```

REFERENCES (Clinical Guidance Used for Threshold Justification)

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